

CUT&Tag and CUT&RUN

Principles, applications and data analysis

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Biological question	Method	Main readout
Which genes change expression?	RNA-seq	RNA abundance
Which regulatory regions are accessible?	ATAC-seq	DNA accessibility
Where does H3K27me3 occur in the genome?	ChIP-seq, CUT&RUN and CUT&Tag	Target-associated DNA enrichment

All three methods map DNA near a selected chromatin target.

Each conventional reaction profiles one target across the genome.

Target	Common association	Biological application	Example publication
H3K4me3	Active or poised promoters	Promoter state during differentiation	scCUT&Tag Bartosovic et al., 2021
H3K27ac	Active promoters and enhancers	Enhancer activity and developmental state	ChIP-seq Creyghton et al., 2010
H3K27me3	Polycomb-associated repression	Repressed domains in small samples / single cells	CUT&Tag Kaya-Okur et al., 2019
H3K9me3	Heterochromatin and repeats	X-chromosome silencing and reactivation	CUT&RUN Ernst et al., 2019
CTCF, TFs or RNA Pol II	Target-specific chromatin association	Factor occupancy or transcriptional regulation	CUT&Tag Kaya-Okur et al., 2019

H3K27me3 means histone H3, lysine 27, trimethylation. A mark suggests a state.
Selected studies shown. Full references and study details are in the speaker notes.

Three routes to a chromatin map

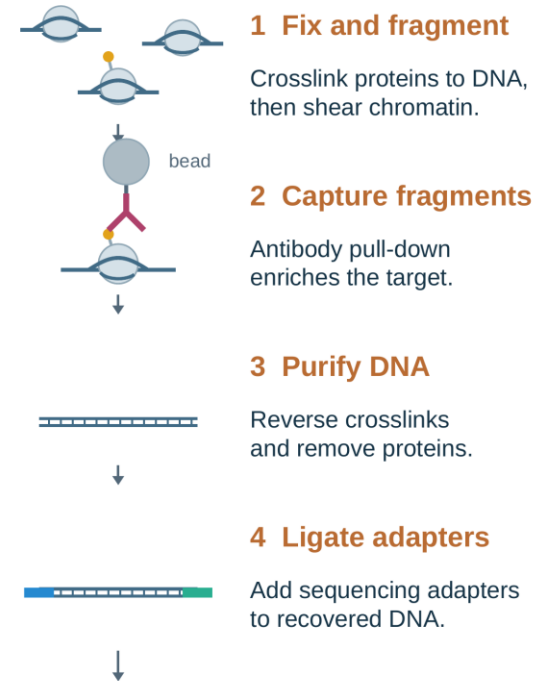
ChIP-seq, CUT&RUN and CUT&Tag

How antibodies select chromatin for sequencing: a histone-mark example

— DNA ● Nucleosome ● Target histone mark Y Antibody — Sequencing adapters Read each method from top to bottom

ChIP-seq

Enrich fragments by pull-down

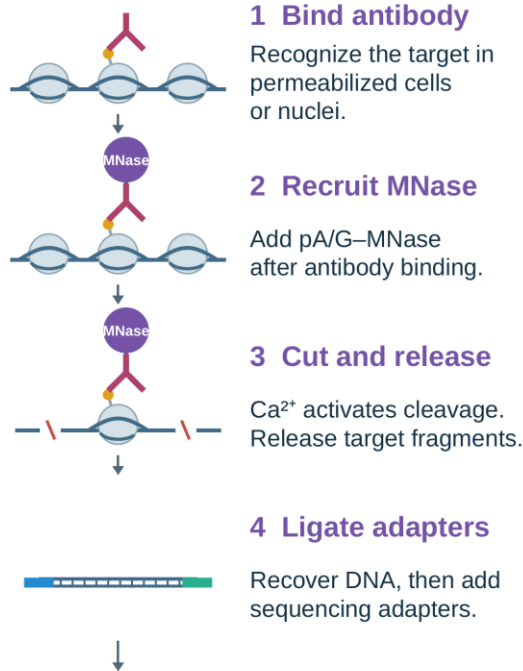


PCR and sequencing

Read alignment reveals genomic regions enriched for the selected target

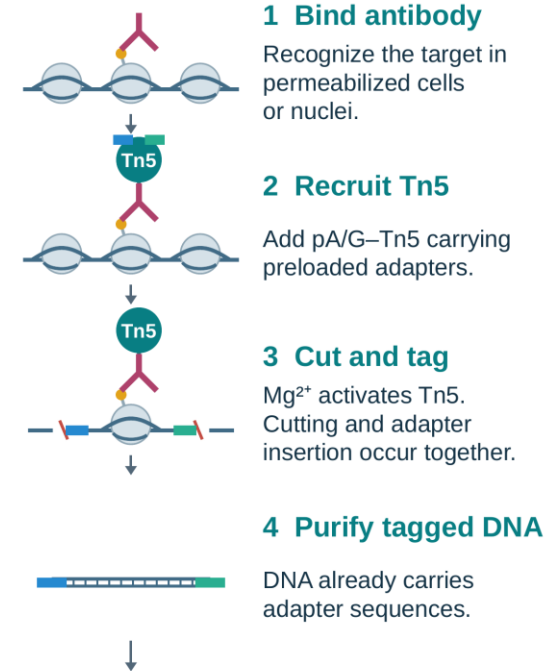
CUT&RUN

Targeted nuclease cleavage



CUT&Tag

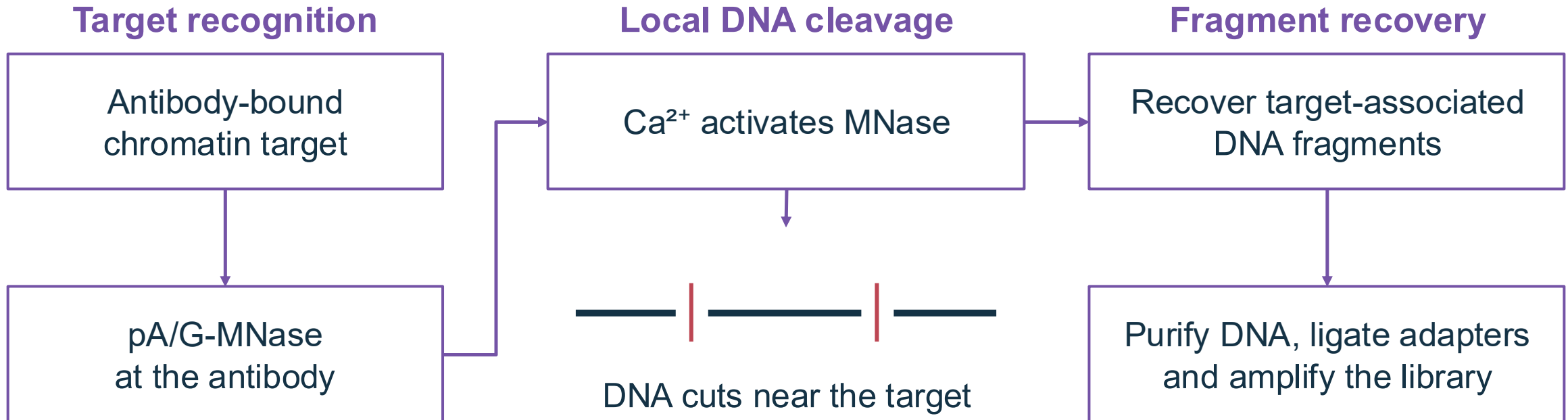
Targeted adapter insertion



Both CUT methods bind antibody before enzyme recruitment.

Simplified conventional workflows. ChIP-seq shows a crosslinked protocol. pA/G = protein A and/or G. Secondary antibodies and washes are omitted.
Sources: Skene & Henikoff (2017), eLife, doi:10.7554/eLife.21856. Kaya-Okur et al. (2019), Nature Communications, doi:10.1038/s41467-019-09982-5.

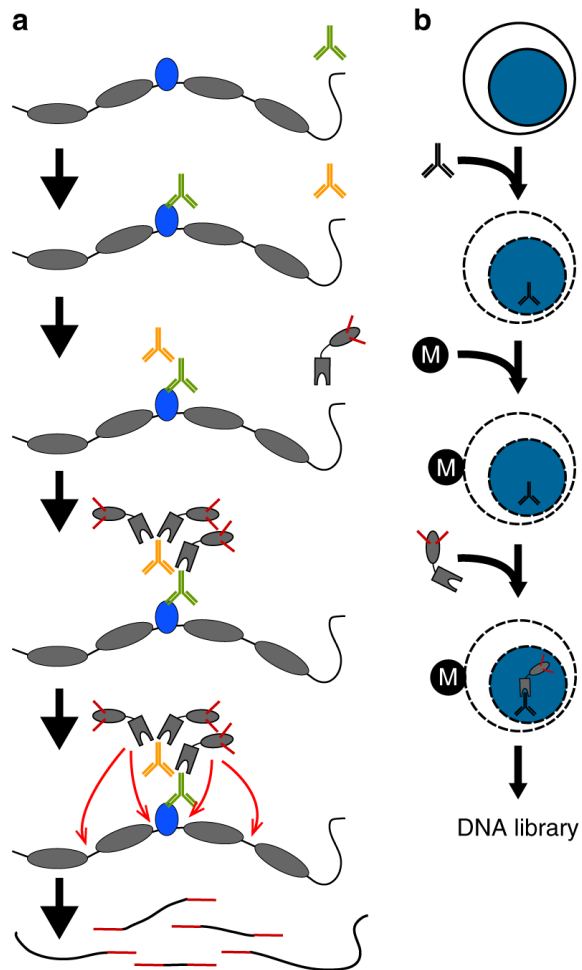
Cleavage Under Targets and Release Using Nuclease



Classic recovery collects released fragments from the supernatant.

Total-DNA and modified recovery protocols also exist.

Cleavage Under Targets and Tagmentation



1 Antibody binding

Permeabilized cells or nuclei expose the target epitope.

2 Secondary antibody

Often increases recruitment of the tethered enzyme.

3 Adapter-loaded Tn5

Protein A or A/G links Tn5 to the antibody complex.

4 Targeted tagmentation

After washing, Mg^{2+} activates cutting and adapter insertion.

5 Sequencing library

DNA recovery, PCR indexing, cleanup and paired-end sequencing.

Feature	ChIP-seq	CUT&RUN	CUT&Tag
Core step	Antibody pull-down of fragmented chromatin	Antibody-directed MNase cleavage	Antibody-directed Tn5 tagmentation
Activation	Fragmentation by sonication or enzyme	Ca ²⁺ activates MNase	Mg ²⁺ activates Tn5
Adapter addition	After DNA recovery	After DNA recovery	During tagmentation
Typical material	Often fixed, fragmented chromatin	Permeabilized cells or nuclei	Permeabilized cells or nuclei
Major concerns	Fixation, solubilization and pull-down	Digestion and fragment recovery	Accessibility and tagmentation conditions
Input / reads	Often more material and sequencing	Often lower input and background	Often lower input and background

Typical workflows shown. Native ChIP and fixed CUT protocols also exist.

Planning example: vehicle and treatment, with 3 independent cultures each.

The number of replicates is an illustration, not a universal sample-size requirement.

Control or design choice	What it helps assess
IgG or no-primary control [2,3]	Nonspecific signal under the same assay conditions
Validated positive-control target [2,3]	Whether the chemistry produces expected enrichment
Target depletion or another validated antibody [1]	Evidence for target specificity
Balanced batches and biological replicates [1]	Technical confounding and reproducibility

Technical sequencing lanes do not replace biological replicates.

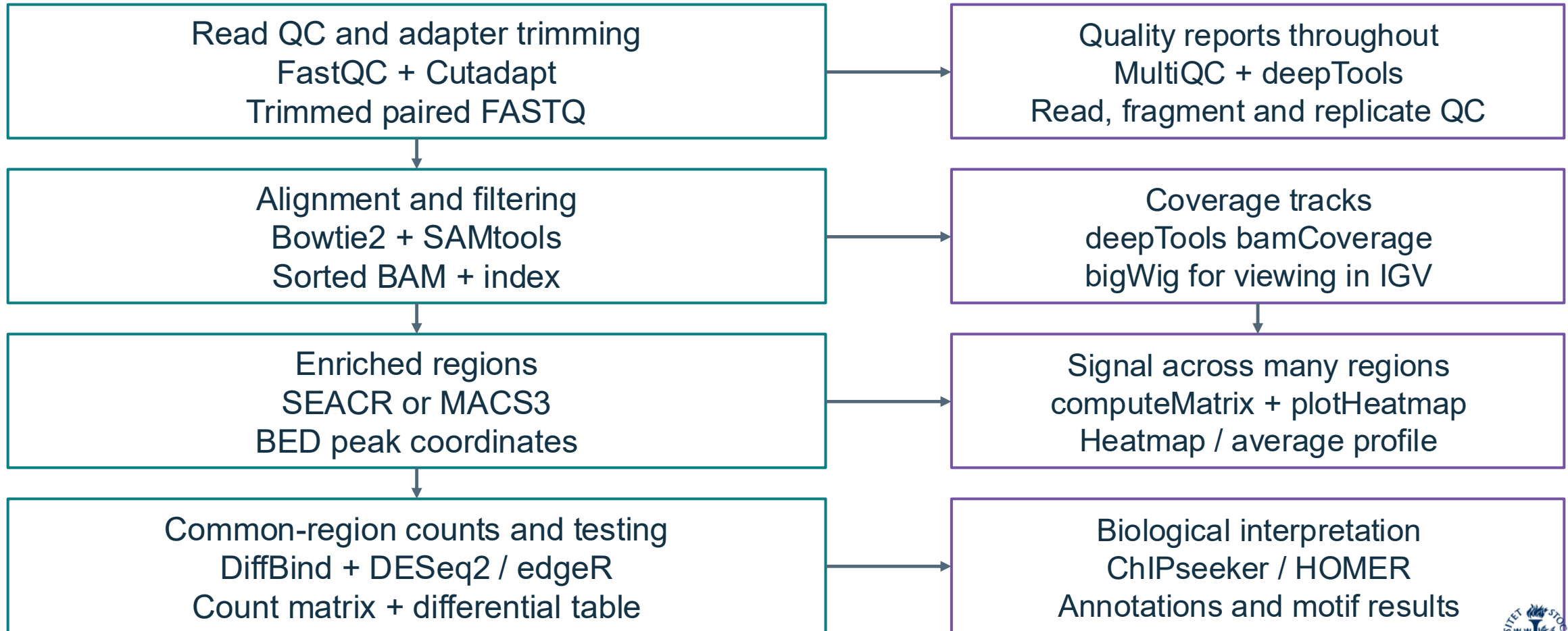
REFERENCES

- [1] Landt SG et al. (2012). Genome Research 22, 1813-1831.
- [2] Skene PJ, Henikoff JG, Henikoff S. (2018).. Nature Protocols 13, 1006-1019.
- [3] Kaya-Okur HS, Janssens DH, Henikoff JG, Ahmad K, Henikoff S. (2020). Nature Protocols 15, 3264-3283.

How the assay changes data analysis

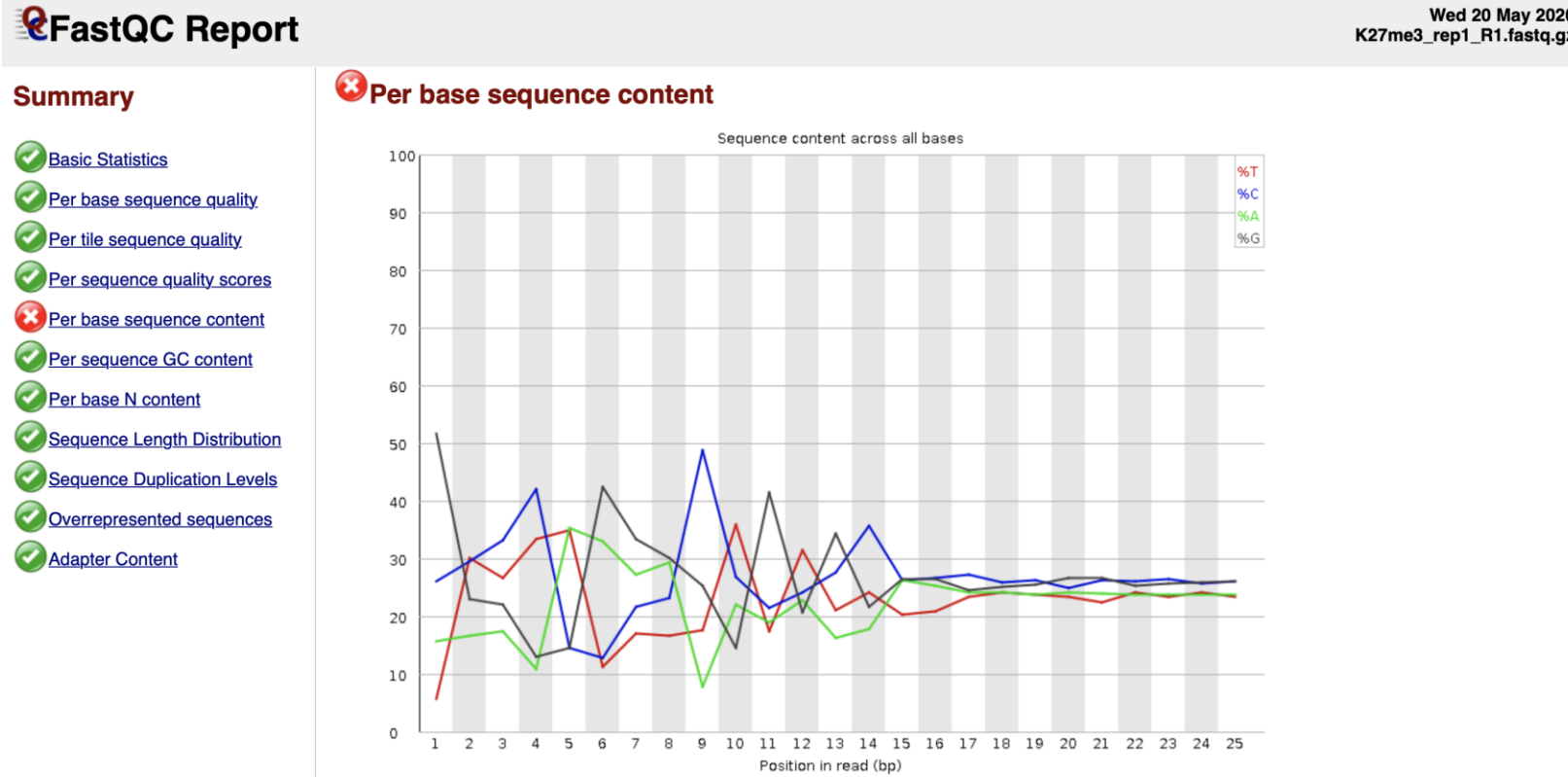
Decision	ChIP-seq	CUT&RUN	CUT&Tag
Fragment pattern	Shearing or enzymatic fragmentation	MNase cuts and fragment recovery	Tn5 insertion near the target
Background control	Matched input and appropriate IP control	Assay-matched IgG or no-primary control	Assay-matched IgG or no-primary control
QC emphasis	Enrichment, complexity and replicate agreement	Short / nucleosomal fragments, recovery and background	Short inserts, adapters and accessible-DNA background
Peak model	Match the target and background profile	Allow sparse signal and target-specific widths	Allow sparse signal and target-specific widths
Shared inference	Common regions, biological replicates, count models and appropriate normalization	Same principles	Same principles

Inputs: paired FASTQ, sample metadata, reference genome and assay controls



SEACR uses fragment bedGraph. MACS3 can use BAMPE. Replicates feed one common region set.

CUT&Tag in K562 cells. Example output from FastQC.



READ THE PLOT

The lines show base proportions, not Phred quality scores.

DO

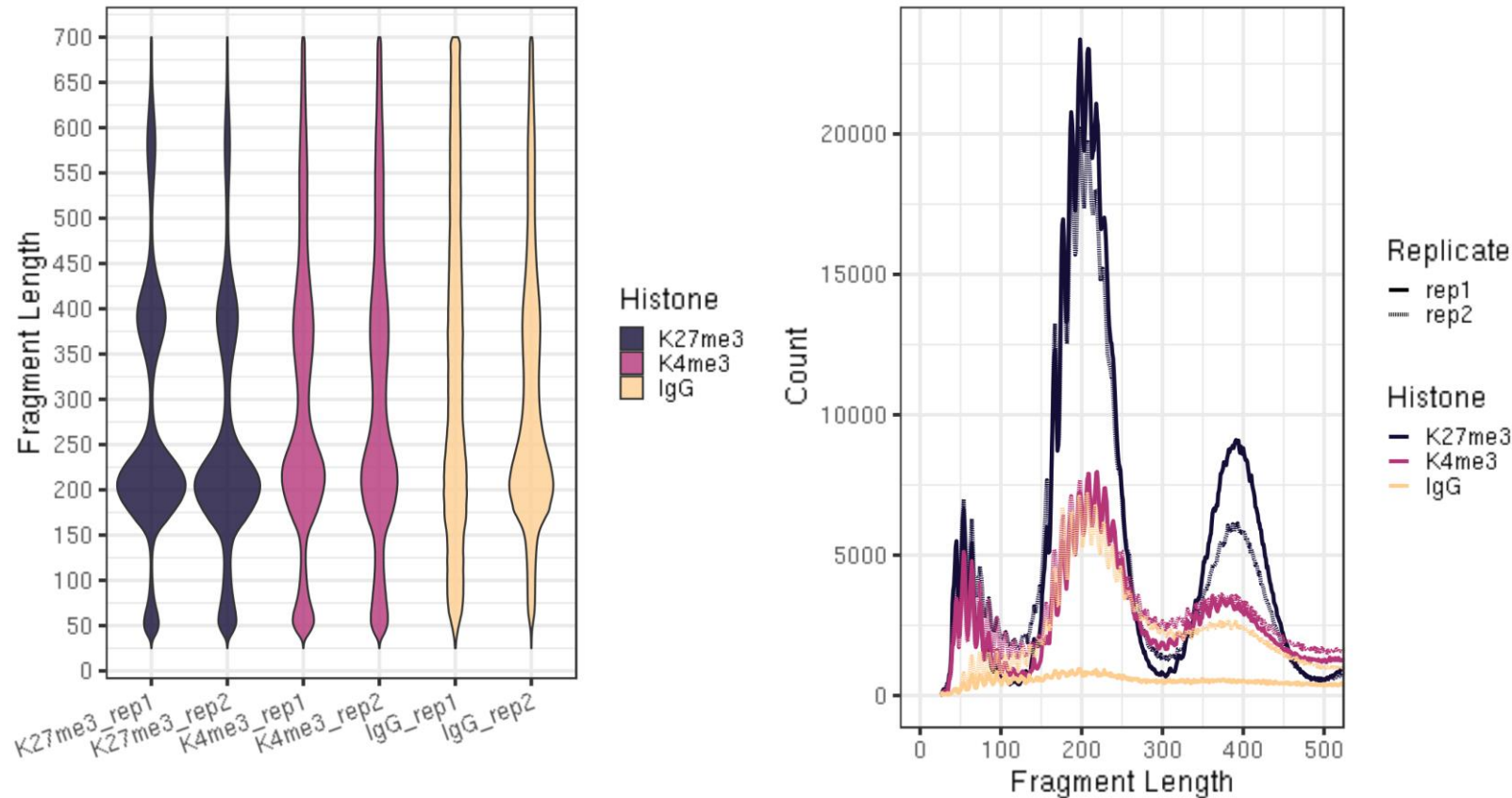
Check base quality and adapters. Trim adapter read-through when present.

DON'T

Reject a library solely because this module flags a sequence bias.

Base composition varies near the read start.

Outputs: fragment-length distribution and duplication metrics. Tools: SAMtools, Picard and R.



READ THE PLOT

Nucleosomal peaks coexist with shorter fragments.

DO

Inspect each replicate and IgG. Record and justify the duplicate policy.

DON'T

Treat every short fragment as noise, or every identical pair as PCR duplication.

Shape and periodicity need target-specific interpretation.

Analyse the short fragments separately. (Henikoff tutorials)

check whether they show reproducible enrichment at the expected target regions

“short” means the **DNA fragment length inferred from paired-end alignment**,
not the length of an individual sequencing read.

CUT&Tag example, we could compare **50–100 bp fragments** with **150–250 bp fragments**

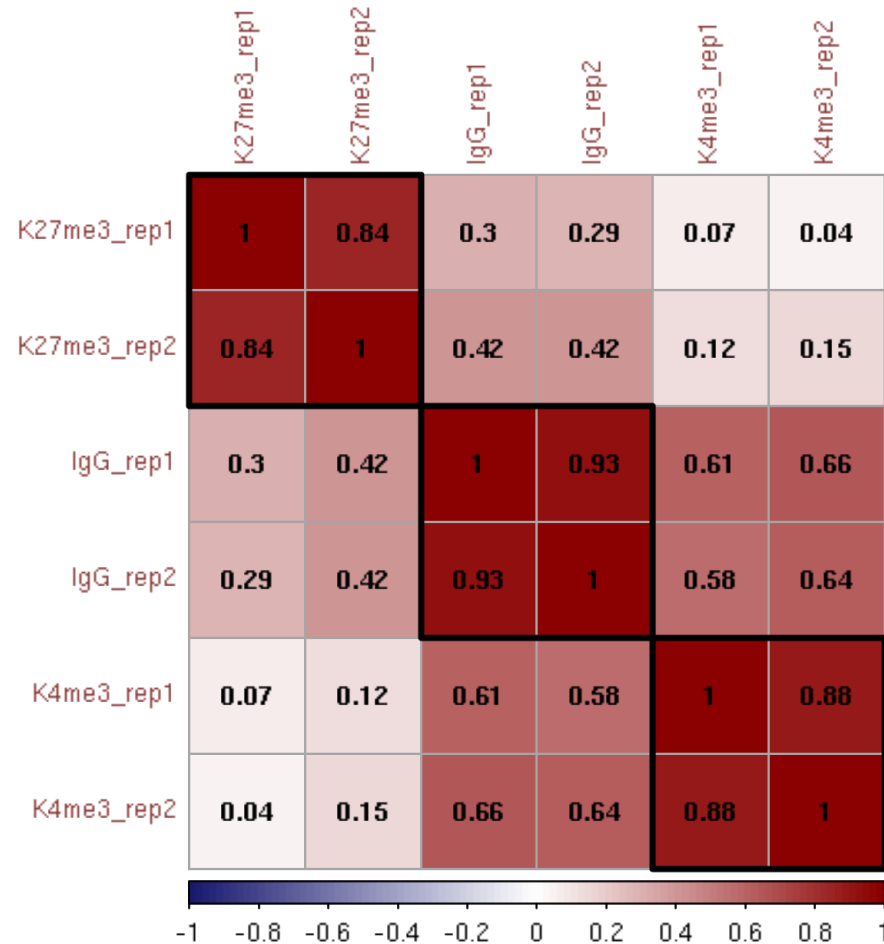
Some methods to evaluate the short fragments

- 1) Are they real genomic fragments?
- 2) Where do they map?
- 3) Are they enriched at known target regions?
- 4) Is the pattern reproducible?

Tools

FastQC, Cutadapt, Bowtie2 and SAMtools
deepTools bamCoverage and IGV
BEDTools intersect
deepTools multiBamSummary and plotCorrelation

Output: sample correlation heatmap.



READ THE PLOT

Replicate correlations: H3K27me3 = 0.84, H3K4me3 = 0.88. IgG also correlates strongly.

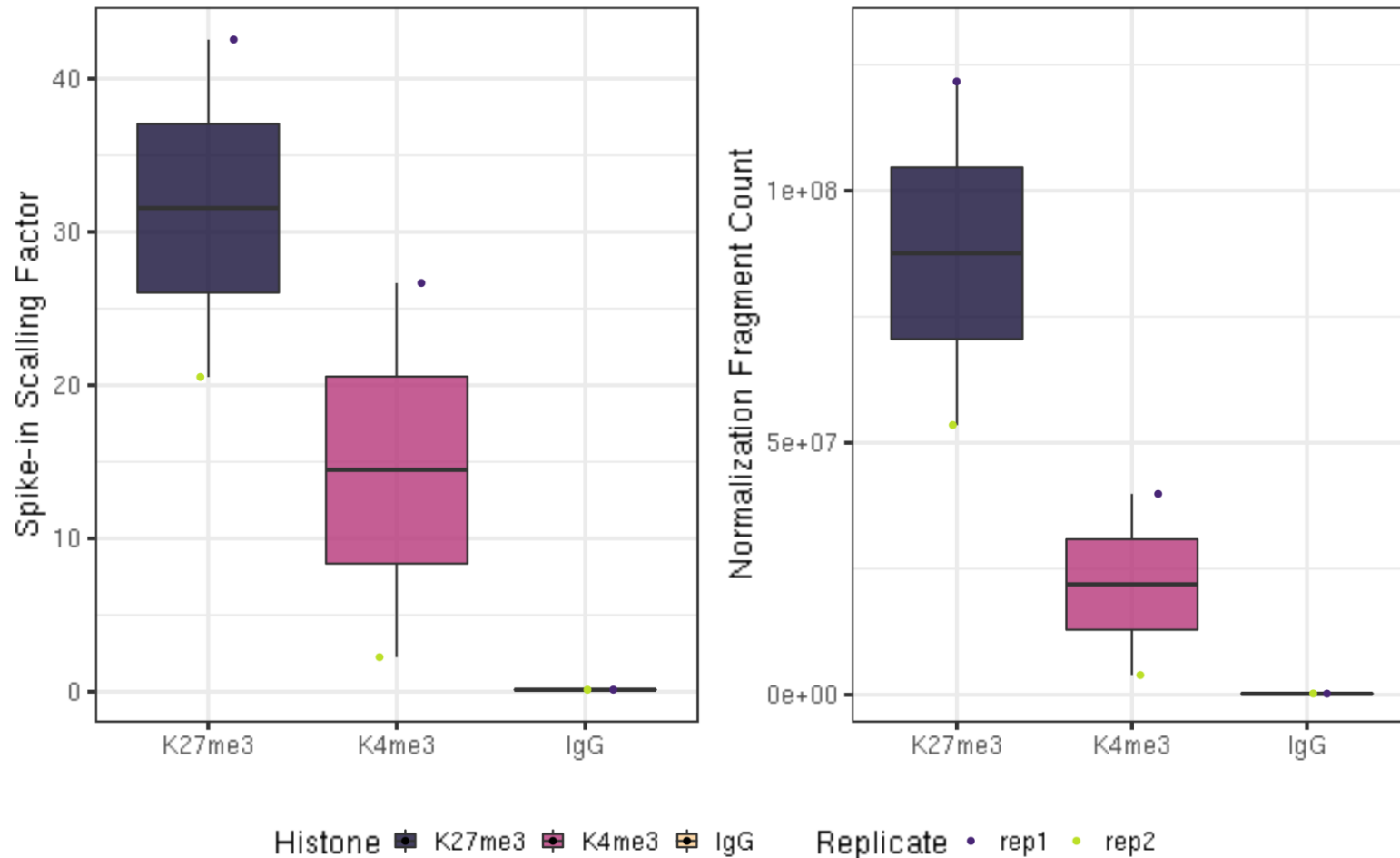
DO

Use the same genomic bins and filtering. State how zero counts and the log transform are handled.

DON'T

Use correlation alone to claim specific enrichment. Inspect background and peak overlap too.

Outputs: scaling factors and calibrated signal. Tools: spike-in alignment, BEDTools and R.



READ THE PLOT

The reference count sets a multiplier for the target coverage.

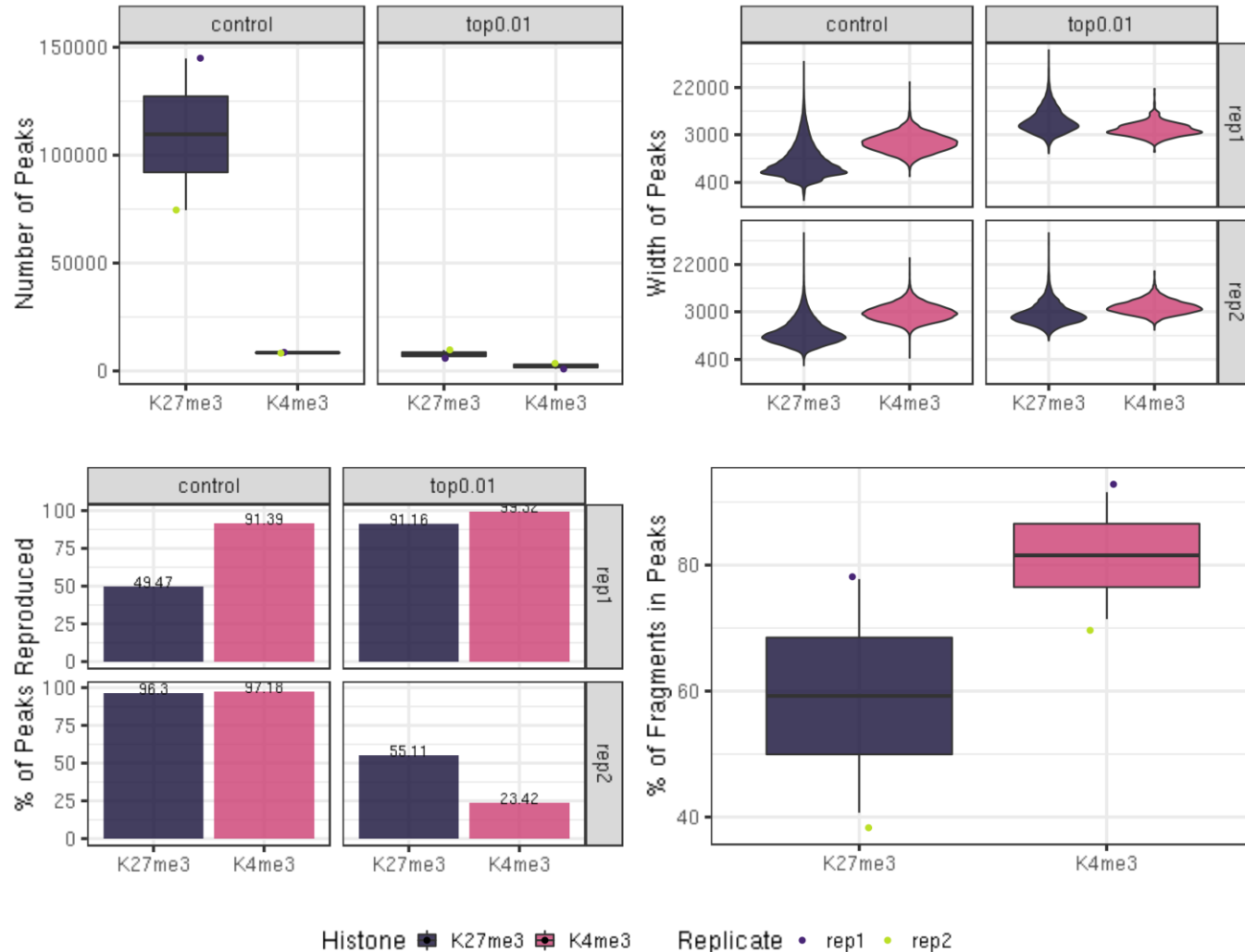
DO

Verify a comparable reference input and protocol across samples.

DON'T

Assume E. coli carry-over is constant across different enzyme preparations.

Outputs: peak count, width, replicate overlap and FRiP. Tools: SEACR, BEDTools and R.



READ THE PLOT

FRiP is the fraction of retained mapped fragments that overlap peaks.

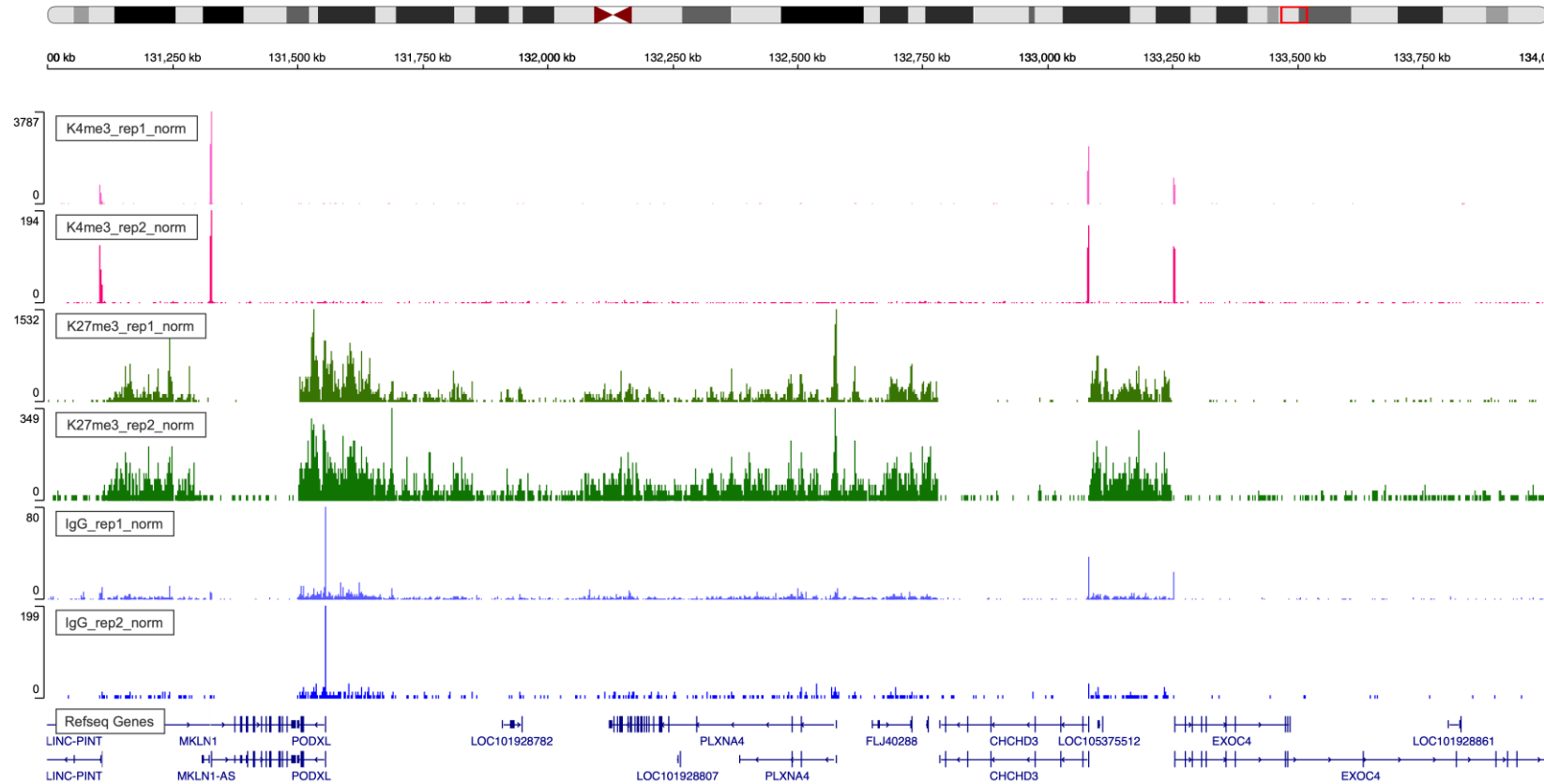
DO

Evaluate IgG, peak width and replicate overlap. Count each fragment once for FRiP.

DON'T

Call top0.01 a 1% FDR. It selects the top 1% of signal blocks by total signal.

Output: normalized bedGraph tracks in IGV. Tutorial region: chr7:131–134 Mb.



DO

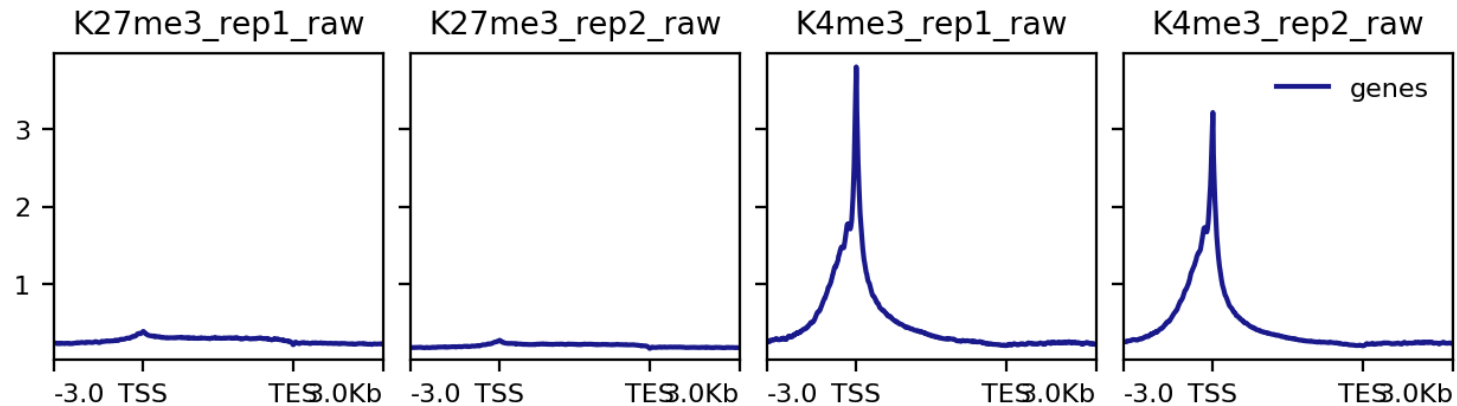
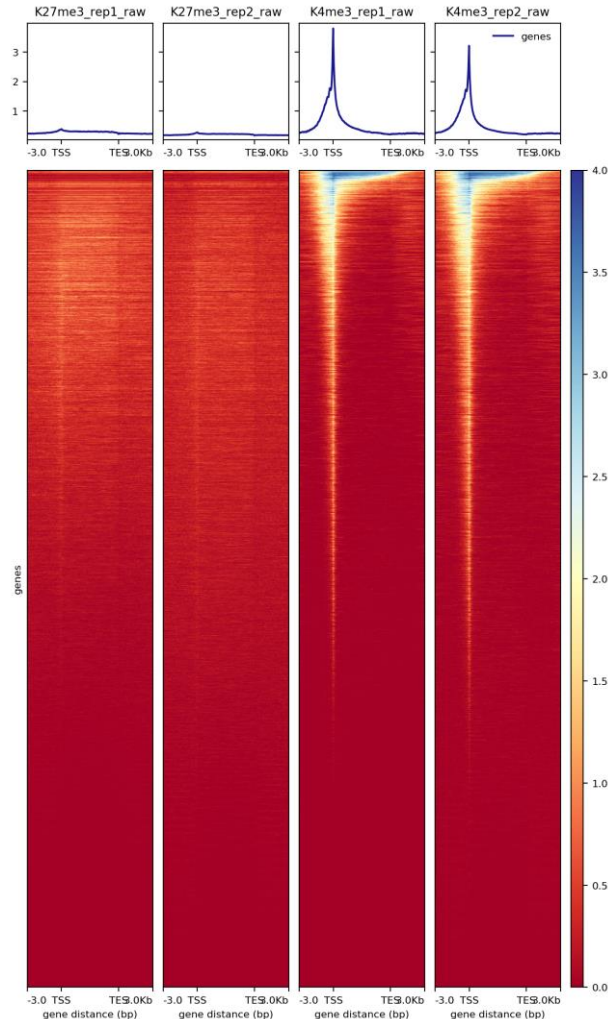
Use a shared y-axis after calibration.

DON'T

Infer abundance from autoscaled peak heights.

Figure: Zheng, Ahmad & Henikoff (2020), §7.1. doi:10.17504/protocols.io.bjk2kkye

Outputs: region-by-position heatmap and mean profile. Tools: deepTools computeMatrix and plotHeatmap.



Rows = genes. Columns = positions across genes and flanks.
H3K4me3 peaks near transcription start sites (TSS).

DO

Keep the same regions, row order and scale when comparing samples.

DON'T

Treat raw coverage or a histone-mark heatmap as gene expression.

Question	Tools	Result to show students
Where are the regions?	ChIPseeker	Annotated region table: promoter / intron / distal, TSS distance and associated gene
Which sequences are enriched?	HOMER	Motif logos and enrichment statistics for candidate transcription factors
Which functions are associated?	clusterProfiler	GO / pathway table or dot plot, adjusted P values and contributing genes

Motif tests need matched background sequences. Gene-set tests need an eligible gene universe.

Annotation supports hypotheses. Gene regulation needs additional evidence.

Existing validated assay or a cross-linked specimen

ChIP-seq can preserve continuity with established datasets.

Native factor occupancy with low background

CUT&RUN is a candidate when the antibody and fragment recovery work well.

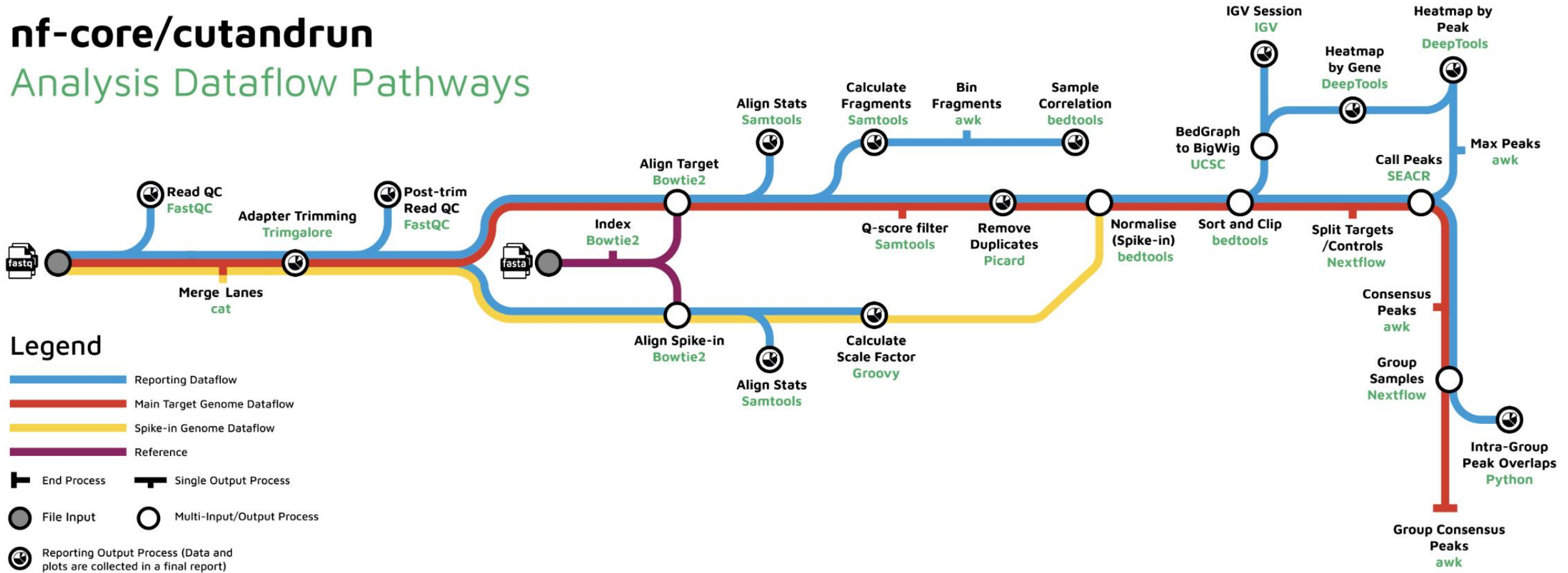
Low-input histone profiling or a single-cell extension

CUT&Tag is a candidate when targeted tagmentation is validated.

Pilot the actual target and specimen. No method is universally best.

nf-core/cutandrun

Analysis Dataflow Pathways



Why researchers choose it

- Efficient profiling from limited material
- Often low background
- Direct adapter insertion

What still needs validation

- Antibody specificity in the assay
- Accessible-DNA background from Tn5
- Target retention during washes

A useful benchmark

A 2025 K562 study reported about 54% average recall of ENCODE peaks for H3K27ac and H3K27me3 across tested CUT&Tag datasets.

Note: Recall means the percentage of ENCODE reference peaks that CUT&Tag also detected

That result is specific to the tested targets, protocols and reference datasets.

ChIP-seq experimental standards

Landt et al. (2012), Genome Research

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3431496/>

CUT&RUN original method

Skene and Henikoff (2017), eLife

<https://doi.org/10.7554/eLife.21856>

CUT&RUN protocol

Skene, Henikoff and Henikoff (2018), Nature Protocols

<https://doi.org/10.1038/nprot.2018.015>

CUT&Tag original method

Kaya-Okur et al. (2019), Nature Communications

<https://doi.org/10.1038/s41467-019-09982-5>

CUT&Tag protocol

Kaya-Okur et al. (2020), Nature Protocols

<https://doi.org/10.1038/s41596-020-0373-x>

Biological application of histone ChIP-seq

Roadmap Epigenomics Consortium (2015), Nature

<https://doi.org/10.1038/nature14248>

Thanks for listening

Questions!!!!